Explicit inconsistency from a derivable sentence and its negation

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof argues by transformation of closed tableaux in prose; no tableau environment occurs in the counted span

Editorial note. The label prop:explicit-inc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed assumption set of a derivable sentence. Gamma proves A iff for some finite B_1,...,B_n in Gamma the set {False A, True B_1,...,True B_n} has a closed tableau.. The hypothesis Gamma proves A is unpacked into a closed witness tableau whose first assumption is False A.
- The True-not rule. From the signed formula True (not A) the rule derives False A.. The inserted line after the assumptions is the conclusion of the True-not rule, restoring the line the replaced assumption used to supply.

Proposition 0.1. *If $\Gamma \vdash A$ and $\neg A \in \Gamma$, then Γ is inconsistent.*

Proof. Suppose $\Gamma \vdash A$ and $\neg A \in \Gamma$. Then there are B_1, ..., $B_n \in \Gamma$ such that

$$\{\bot(A), \top(B_1), \ldots, \top(B_n)\}$$

has a closed tableau. Replace the assumption $\bot(A)$ by $\top(\neg A)$, and insert the conclusion of $\top\neg$ applied to $\bot(A)$ after the assumptions. Any sentence in the tableau justified by appeal to line 1 in the old tableau is now justified by appeal to line $n + 1$. So if the old tableau was closed, the new one is. It shows that Γ is inconsistent, since all assumptions are in Γ. □